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(1) Research Program/Project/Study Title: PREFERENCE OF CHEWABLE INDIAN MANGO ROLL WITH CHILI	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	NA

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	PILAR
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(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Food, Small Scale Entrepreneurs, housewives,	(8) Discipline/Field of Specialization Food/Natural Science
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ABSTRACT

The importance of fruit in human diet is universally recognized. Fruits are not only ready made delicious food but also valued for vitamin and mineral contents. Mangoes in the Philippines are very abundant and in fact, it is sometimes free during mango season. There are various mango trees that you can find in farms and in the neighborhood. It is one of the most favorite fruit of Filipinos. This study was conducted to determine the acceptability and preference of Indian Mango roll added with herbal condiments. Specifically aims to determine the significant difference of Indian mango roll with chili in terms of flavor, color, texture and general acceptability in varying proportions.

To obtain the answer of the hypothesis the following procedure was followed selecting good quality of Indian mango and herbal condiments, then preparing the treatment for experimentation up to the evaluation by the panelist. After the evaluation the data was gathered by the researcher for statistical analysis using mean and ANOVA. The result showed that Only Treatment A and D chewable Indian mango roll with chili are acceptable and like slightly in terms of flavor, color and texture by the panel of consumers who tested the said product. The hypothesis was rejected since there is a significant difference among the treatments with varying proportions of chewable Indian mango roll added with chili.

Keywords: Indian Mango Roll, Chili

Introduction

Indian mango is a heart shape mango originated in India but well planted in the Philippines, it has 3 stages of ripening: 1. Unripe or premature – commonly sour taste but other varieties are not, it is best with dip or condiments, 2. Almost ripe – this mango fruit stage is the most favorite of Filipinos, the sweetness and sour combination of its taste is really delicious, irresistible and you can consume it raw without sourness, and 3. Ripe–ripe Indian mangoes are rarely picked, due to the softness of texture and unique strong sweetness that most of people do not like. But in typical Mangoes, ripe is the best time to consume, only Indian mangoes is very different when it comes to ripening. Those 3 stages of mangoes has different taste, texture and smell. Different people has different taste so it will depends on you on what stage of Indian mango fruit you will like.

Philippines is one of the exporter of Mangoes in Asia. From dried to fresh mangoes, Philippines always contributing in mango Industry and it has a high quality. Mangoes in the Philippines is very abundant and in fact, it can be free when mango season arrives. There's a lot of mango trees that you can find and there are also farms that has planted mangoes. You can always find mangoes even on street and mangoes is the most favorite fruit of Filipinos.

Indian mangoes is very rich in nutrients like Vitamin A, Folate, Vitamin C, Vitamin E, Potassium, Calcium, Iron, Magnesium, Phosphorus, Copper, Selenium, Manganese and Zinc.

The importance of fruit in human diet is universally recognized. Fruits are not only readymade delicious food but there is also valued for their vitamin and mineral contents without these the human cannot continue to be healthy or resistant to disease. Mango is good source of vitamin A and C, minerals and calories. As they are not cooked, provide a high nutrition value. This fruit is seasonal, Mango often forms a glut in the market during harvesting time. If this fruit is processed into candy at the grower's level, they can sale it at good price and consumers will have the taste in off season. Mango is perishable and cannot be stored for long time for its inherent compositional and textural characteristics. Being highly perishable in nature, larger portion of harvested fruits goes to wastage due to lack of proper postharvest handling and storage technique. Due to seasonal abundance and short market life, sometimes their price become low and a considerable amount is lost. But in this perishable fruit is processed into shelf-stable products, the above losses can be minimized great extent. (Molla, MM).

Candies are palatable food item, composed of sweet taste and high calorie. Processing of fruits into candies is an osmotic dehydration process where sugar goes into the tissues bringing out water into the syrup. As

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sugar is transferred from the syrup to the tissues, the concentration goes up inside the fruits while moisture content becomes lower and lower. A fruit impregnated with sugar and glucose, and subsequently drained and dried is called candied fruit (Girdhari, 1986).

Hence, the researchers conducted a systematic study considering that the abundance and availability of the Indian Mango actually surrounds the vicinity and see how it is blended herbal condiments in the absence of sugar.

Statement of the Problem

This study was conducted to determine the general view and acceptability level of consumers very particular on the chewable indian mango roll.

Specifically this study has the following objectives:

1. Preferences of Chewable Indian Mango Roll with Chili in terms of
 - a. Flavor
 - b. Color
 - c. Texture
 - d. General Acceptability
2. There is no significant difference on the different treatment of chewable indian mango roll with chili

MATERIALS AND EQUIPMENT

The following materials and equipment were used in the study:

Strainer, Bucket, Knife, Chopping Board, Measuring Cup, Measuring Spoon, Spatula, stove, pot, apron, hairnet, pot holder, wooden ladle, drying table, banana leaves, containers, mixing bowls, grinder/blender
Ingredients: Ripe Indian Mango, Chili.

INGREDIENTS

Ripe Indian Mango, Chili

METHODS OF INVESTIGATION

This study used the experimental method of investigation and the product was evaluated by selected panelist using the hedonic scale as a score sheets provided to them with corresponding scale.

PANELIST OF THE STUDY

The panelist were 30 consumers from CapSu Pilar. The data gathered were subjected to statistical analysis.

TREATMENT

Mango Roll with Chilui

- Treatment A – (100% indian mango/10 % chili)
- Treatment B – (100 %indian mango/15 % chili)
- Treatment C – (100 %indian mango/20 % chili)
- Treatment D – 100% indian mango

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PROCEDURE

Preparation

Select a good quality of ripe Indian mango. Wash the indian mango thoroughly with running water to eliminate dirt and germs. Slice the indian mango and scoop the pulp. Puree the pulp using blender. After blending, heat over low fire to eliminate excess water. After the desired thickness weighting was done and chili was mixed according to its treatment. Product were scoop, put in banana leaf and distribute evenly into desired thickness. It was sun dried. After its desired dryness it was evenly sliced and packed.

Preparation of Condiments

Chili

Picking and select red/green chili. Wash to eliminate dirt. Slice into small pieces. Sun dry and grind into powder.

SCORING AND CATEGORIZATION OF VARIABLES

The mango roll was rated by the panelist according to quality such as, flavor, texture, color and general acceptability. Each of this qualities was assigned with corresponding weight and score were categorized as follows:

Sensory Quality	Wright	Range of Score
Like Extremely	9	8.12-9.00
Like very much	8	7.23-8.11
Like Moderately	7	6.34-7.22
Like Slightly	6	5.45-6.33
Neither like nor dislike	5	4.56-5.44
Slightly Dislike	4	3.67-4.55
Moderately Dislike	3	2.78-3.66
Very Much Dislike	2	1.89-2.77
Extremely Dislike	1	1.00-1.88

DATA GATHERING PROCEDURE

The data were gathered using the score sheet on hedonic scale. The four treatments were presented for testing and evaluation by the 30 consumers. The panelist was briefed before about the product was evaluated as to its flavor, color, texture and general acceptability.

ANALYSIS OF DATA

To determine the acceptability of the product, the evaluators were guided in their evaluation using a score sheet and were free to express their judgment.

The panelist rating were analyzed using Complete Randomized Design (CRD). The gathering of data was interpreted through mean. Analysis of variance was further tested using F-test.

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RESULTS AND DISCUSSION

Chewable Indian Mango with Chili

Flavor

Table 1 presents the acceptability of the chewable indian mango roll with chilli as tested/evaluated by the panel of consumers in terms of flavor. Treatment A with a mean of 6.60 were like moderately and treatment D with a mean of 6.10 were like slightly. Treatment B and C were both rated dislike by the tasters with a mean of 4.71 and 5.20 respectively. Based from the result we can say that only treatment C and Treatment D were acceptable by the consumers when it comes to flavor.

Table 1. Level of Acceptability in terms of Flavor.

TREATMENTS	M	SD	VERBAL INTERPRETATION
Treatment A	6.60	1.54	LIKE MODERATELY
Treatment B	4.70	0.60	DISLIKE
Treatment C	5.20	1.19	DISLIKE
Treatment D	6.10	1.37	LIKE SLIGHTLY
TOTAL	5.65	1.42	LIKE SLIGHTLY

Note: * $p < 0.05$.

Color

Table 2 presents the acceptability of the chewable indian mango roll with chilli as tested/evaluated by the panel of consumers in terms of color. In terms of color it is like moderately by the respondents with a grand mean of 6.48. Treatment A were like very much ($M=7.70$, $SD=1.05$), Treatment D were also like very much ($M=7.23$, $SD=1.10$). Treatment C were like slightly ($M=5.80$, $SD=1.79$) and Treatment B ($M=5.20$, $SD=1.16$) were dislike by the panel of tasters. Based from the result we can say that Treatment A, D and C chewable indian mango roll with chilli are acceptable in terms of color as evaluated by the panel of consumers who tested the said product.

Table 2. Level of Acceptability in terms of color.

TREATMENTS	M	SD	VERBAL INTERPRETATION
Treatment A	7.70	1.05	LIKE VERY MUCH
Treatment B	5.20	1.16	DISLIKE
Treatment C	5.80	1.79	LIKE SLIGHTLY
Treatment D	7.23	1.10	LIKE VERY MUCH
TOTAL	6.48	1.65	LIKE MODERATELY

Note: * $p < 0.05$.

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Texture

Table 3 presents the acceptability of the chewable indian mango roll with chilli as tested/evaluated by the panel of consumers in terms of texture. In terms of texture it shows that treatment D is like moderately by the participants who evaluated the chewable indian mango roll ($n=30$, $m=6.87$, $SD=1.01$), followed by treatment A ($n=30$, $m=7.00$, $SD=1.08$) which is liked moderately, then treatment C ($n=30$, $m=5.53$, $SD=1.25$) which is like slightly and B ($n=30$, $m=5.03$, $SD=1.03$) which is also disliked by the panel of tasters. Based from the result we can say that only treatment D, A and C chewable indian mango with chilli are acceptable in terms of texture and like moderately by the panel of consumers who tested the said product.

Table 3. Level of Acceptability in terms of texture.

TREATMENTS	<i>M</i>	<i>SD</i>	VERBAL INTERPRETATION
Treatment A	7.00	1.08	LIKE MODERATELY
Treatment B	5.03	1.03	DISLIKE
Treatment C	5.53	1.25	LIKE SLIGHTLY
Treatment D	6.87	1.01	LIKE MODERATELY
TOTAL	6.10	1.38	LIKE SLIGHTLY

Note: $*p < 0.05$

General Acceptability

Data in table 4 revealed the over all acceptability of the chewable indian mango roll with chilli as varied proportions as tested/evaluated by the panel of consumers. It shows that the chewable indian mango roll with chilli with a mean of 5.97 is reasonably interpreted as like slightly as evaluated and tested by the panel of consumers. Specifically, treatment A ($n=30$, $m=6.77$, $SD=1.10$) is like slightly, followed by treatment D ($n=30$, $m=6.60$, $SD=0.93$) which is considered as acceptable and is like slightly also, then treatment C ($n=30$, $m=5.43$, $SD=1.04$) and lastly treatment B ($n=30$, $m=5.07$, $SD=1.14$) who got the lowest mean is disliked by the panelist. Based from the result we can say that only Treatment A and D chewable indian mango roll with chili are acceptable and like slightly by the panel of consumers who tested the said product

Table 4. Level of Acceptability in terms of general Acceptability

TREATMENTS	<i>M</i>	<i>SD</i>	VERBAL INTERPRETATION
Treatment A	6.77	1.10	LIKE MODERATELY
Treatment B	5.07	1.14	DISLIKE
Treatment C	5.43	1.04	DISLIKE
Treatment D	6.60	0.93	LIKE MODERATELY
TOTAL	5.97	1.28	LIKE SLIGHTLY

Note: $*p < 0.05$.

Table 5 presents the analyzed result from the one-way ANOVA in the experimentation using varying proportions of chewable indian mango roll with chilli. The hypothesis stated that there is no significant difference between the chewable indian mango roll in terms of flavor, color and texture and general acceptability was not

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accepted due to the significant p -value .000. This means that there is a significant difference among the treatments of chewable indian mango roll with chilli.

Table 5. One-Way ANOVA (Experimental Group)

		SUM OF SQUARES	DF	MEAN SQUARE	F	SIG.
FLAVOR	Between Groups	66.300	3	22.100	14.649	.000
	Within Groups	175.000	116	1.509		
	Total	241.300	119			
COLOR	Between Groups	124.700	3	41.567	24.197	.000
	Within Groups	199.267	116	1.718		
	Total	323.967	119			
TEXTURES	Between Groups	85.692	3	28.564	23.684	.000
	Within Groups	139.900	116	1.206		
	Total	225.592	119			
ACCEPTABILITY	Between Groups	64.067	3	21.356	19.085	.000
	Within Groups	129.800	116	1.119		
	Total	193.867	119			

Note: $*p < 0.05$.

The data revealed as shown in table 6 presents the Post hoc analysis (p-values for pairwise t-Test) for the chewable indian mango roll with chilli in varying proportions in terms of acceptability, flavor, color and texture.

Table 6. Post hoc analysis (p-values for pairwise t-Tests)

	FLAVOR				COLOR				TEXTURE				ACCEPTABILITY			
Treatment	T _A	T _B	T _C	T _D	T _A	T _B	T _C	T _D	T _A	T _B	T _C	T _D	T _A	T _B	T _C	T _D
T _A		.000	.000	.706		.000	.000	1.000		.000	.000	1.000		.000	.000	1.000
T _B	.000		.706	.000	.000		.473	.000	.000		.483	.000	.000		1.000	.000
T _C	.000	.706		.032	.000	.473		.000	.000	.483		.000	.000	1.000		.000
T _D	.706	.000	.032		1.000	.000	.000		1.000	.000	.000		1.000	.000	.000	

Note: $*p < 0.05$.

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Conclusions

Based on the results of the study, the following conclusions were drawn:

1. The treatment with zero condiments and treatment with 10% chili were acceptable in terms of flavor, color, texture and general acceptability
2. There is a significant difference in the different treatments in terms of flavor, color, texture and general acceptability

Recommendations

Based on the results of the study, the following recommendations were made:

1. Use the treatment A (with 10% chili) proportion of chilli in making chewable indian mango roll and treatment D (no chili added).
2. Conduct a study in formulating other product of indian mango using other herbal condiments.
3. Conduct a study on shelf life of indian mango roll product
4. Further improvement of research methodology of preparing the number of materials for the varying proportions of the treatment.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.) **Output of the Study**

Chewable Indian Mango Roll fortified with various Herbal Condements

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

Demo Activity for extension servives

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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